



## **NATIONAL SERVICE SCHEME**

### **21st AISSMS Engineering Today 2026 Technical Symposium and Exposition**

### **URJA 2026 – Energy Innovation Working Model Competition Rulebook**

**Innovate • Build • Demonstrate**

**SEPTEMBER 30, 2026**

#### **1. ABOUT URJA 2026**

URJA 2026 is an open-ended working model competition focused on renewable, sustainable and efficient use of energy. Participants are expected to identify an energy-related challenge and develop an original, functional and demonstrable working model.

URJA is not limited to electricity generation. Participants may explore energy generation, conservation, efficiency, conversion, storage, recovery, utilization and smart management. Models may involve electrical, thermal, mechanical, chemical, kinetic or other renewable/recoverable forms of energy.

#### **2. SCOPE OF INNOVATION**

Participants may explore, but are not limited to:

- Renewable energy generation and utilization.
- Energy conservation and reduction of energy wastage.
- Energy efficiency and improved utilization of existing energy.
- Energy storage and management.
- Energy conversion and recovery from otherwise wasted energy.
- Smart energy systems, monitoring, automation and intelligent control.
- Sustainable energy solutions for homes, communities, agriculture, rural and remote areas.
- Any other original energy-related innovation.

These are broad directions and are not fixed problem statements. Participants are free to identify their own challenge and solution.

#### **3. ELIGIBILITY AND TEAM FORMATION**

1. The competition is open to all students.
2. A team may consist of 2 to 4 participants.
3. Each team must nominate one participant as the primary presenter.
4. The registered team must develop and present the submitted model.

#### **4. WORKING MODEL REQUIREMENTS**

1. A working model is mandatory for participation.
2. The model must demonstrate its intended working principle during the event.



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3. The model should address a clearly identifiable energy-related problem, need or opportunity.
4. The team must be able to explain the source/form of energy, working principle, innovation and intended application.
5. The model must be transported, assembled and operated by the participating team.
6. The model should be safe to operate at the venue.
7. Teams may use commercially available components, sensors, motors, controllers or other standard parts as part of their own implementation.

## 5. PRESENTATION AND DEMONSTRATION

1. Each team will get a maximum of 10 minutes for presentation and demonstration.
2. Teams should explain the problem, proposed solution, working principle, innovation and practical application.
3. The working model must be demonstrated within the allotted time.
4. The remaining time may be used for questions from the judges.
5. Teams should be prepared to answer questions related to feasibility, limitations, scalability, sustainability and real-world application.

## 6. ORIGINALITY AND ACADEMIC INTEGRITY

1. The core idea and implementation presented by the team should be its own work.
2. Teams may use external references, existing technologies and commercially available components for learning and implementation.
3. A copied model, project or concept must not be presented as an original innovation.
4. If a significant external technology or pre-existing system is used, the team should clearly acknowledge it when asked by the judges.
5. Teams may be asked to explain the design and working of their model to establish their understanding.

## 7. EVALUATION CRITERIA

The models will be evaluated on:

- Working Principle
- Innovation
- Working Model / Functionality
- Practicality and Feasibility
- Sustainability / Energy Impact
- Presentation and Question & Answer



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### 8. REGISTRATION AND PARTICIPATION

1. Registration fee: ₹100/- per participant.
2. Registration must be completed through the official registration process.
3. Participants must bring their registered working model to the venue on 30 September 2026.
4. Participation certificates will be provided to all registered participants.
5. Prizes worth ₹8,000/- will be awarded as per the organizers' decision.

### 9. GENERAL RULES

1. Participants must report to the venue and complete setup within the time communicated by the organizers.
2. Teams are responsible for their own model, components and equipment.
3. The organizers will provide only the facilities specifically communicated before the event.
4. Teams should not cause damage to the venue or equipment.
5. Any change in event procedure, timing or arrangements will be communicated by the organizers.
6. The decision of the judges and organizers regarding the final evaluation will be treated as final.

**NO FIXED PROBLEM. NO FIXED TECHNOLOGY. JUST YOUR ENERGY INNOVATION.**